Group Coursework Submission Form

Specialist Masters Programme

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MSc Actuarial Science 2024-25
SMM048 Insurance Risk Modelling (Subject CS2)
Group Coursework 2

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1 Introduction

An insurance company underwrote 500 policies on 01/01/2024, and the claims for each policy are recorded in an Excel spreadsheet. The claims are provided in columns named `dataset1` to `dataset14`, with the data for this analysis coming from **dataset7**. It is important to note that each policy raised exactly one claim during the year 2024.

As the first step in the exploratory data analysis, we compute the statistical measures of the dataset7, which are presented in Table.

Statistic	**Value**
Minimum	1.0000
1st Quartile	47.7650
Median	160.8950
Mean	553.1145
3rd Quartile	482.3450
Maximum	24373.8300

Table 1: Summary Statistics of Dataset 7

We then visualize the data graphically with a boxplot and histogram, as displayed in Figure 1.

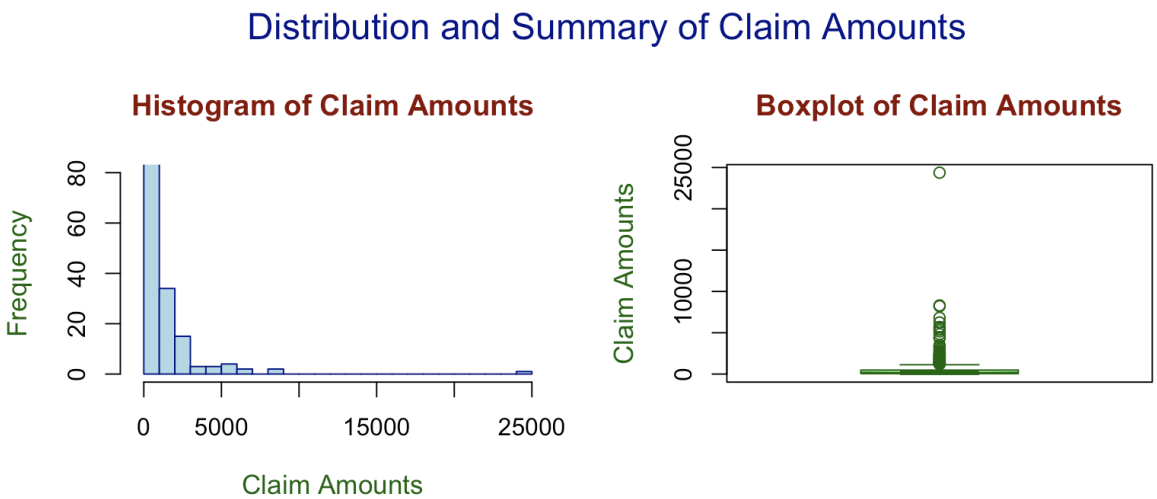

Figure 1: Boxplot and Histogram for the claim amounts

From the histogram and summary statistics, it is evident that the majority of the claims are concentrated between 0 and 5,000. Furthermore, the distance between the third quartile and the median, compared to the distance between the first quartile and the median, suggests a significant right skewness in the distribution, which is further confirmed by the histogram. Additionally, the boxplot reveals the presence of potential outliers, which will be analysed later to assess their impact on the estimation process.

The goal of this research is to estimate the pure premium using various statistical methods, including non-parametric, bootstrap, and parametric techniques, followed by an analysis of combined data.

2 Task A1: Non-Parametric Estimation for Pure Premium

In this Task, we calculate the **Pure Premium Estimator**, a key metric used to estimate the cost of insurance based on historical claims data. The formula for the pure premium estimator is as follows:

$$\text{Pure } \hat{\text{Premium}} = 0.65 \times \bar{x} + 0.20 \times \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} + 0.15 \times \hat{Q_{90\%}}(X) \tag{1}$$

Where:

- $\bar{x}$ is the sample mean, representing the average claim size.
- $\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$ is the sample standard deviation.
- $\hat{Q_{90\%}}(X)$ is the 90th percentile of the claim size data, which accounts for extreme claims.

The formula essentially combines the average claim size, the variability of the claims (through the standard deviation), and the impact of large claims (through the 90th percentile) to provide a comprehensive estimate of the pure premium.

The pure premium estimator is calculated by first determining the sample mean, sample standard deviation, and the 90th percentile of the claim size data. After running the R code, the non-parametric estimate of the pure premium was obtained as:

Non-Parametric Estimate of Pure Premium: 856.9033

This non-parametric estimate of the pure premium effectively incorporates both the central tendency and the variability of the claims, providing a reliable estimate that accounts for extreme values in the dataset7.

3 Task A2: Non-Parametric Bootstrap Sampling for Estimation

In this task, we redo the estimations from Task A1 using **non-parametric bootstrap sampling**, a resampling technique that does not rely on assumptions about the underlying data distribution. This method allows us to estimate the distribution of the pure premium in a data-driven manner, improving the robustness of the estimation.

To perform the non-parametric bootstrap, we resampled the original data with replacement to create multiple bootstrap samples. Each sample was then used to estimate the pure premium using Equation (1). The process was repeated for various bootstrap sample sizes: 3,000, 5,000, 10,000, and 20,000 repetitions. The performance of these estimates was assessed using the **Mean Squared Error (MSE)** and **Variance (VAR)**. Selecting an appropriate bootstrap sample size involves balancing computational efficiency and estimation accuracy.

Bootstrap Size	MSE	Variance	Difference (MSE - VAR)
3,000	16,178	16,184	-5.394
5,000	16,473	16,476	-3.295
10,000	16,706	16,707	-1.671
20,000	16,723	16,724	-0.836

Table 2: Mean Squared Error (MSE) and Variance for Different Bootstrap Sample Sizes

The results reveal a clear trend: as the number of bootstrap repetitions increases, both MSE and Variance exhibit slight growth, while the difference (MSE - Variance) steadily decreases, indicating that the bias correction stabilizes with larger sample sizes. Smaller sample sizes (3,000, 5,000) are computationally efficient but yield higher MSE and Variance, making the estimates less reliable. In contrast, larger sample sizes (10,000, 20,000) provide more stable and accurate estimates but come with increased computational costs. Our analysis indicates that 10,000 bootstrap samples is the optimal choice progressing to stability.

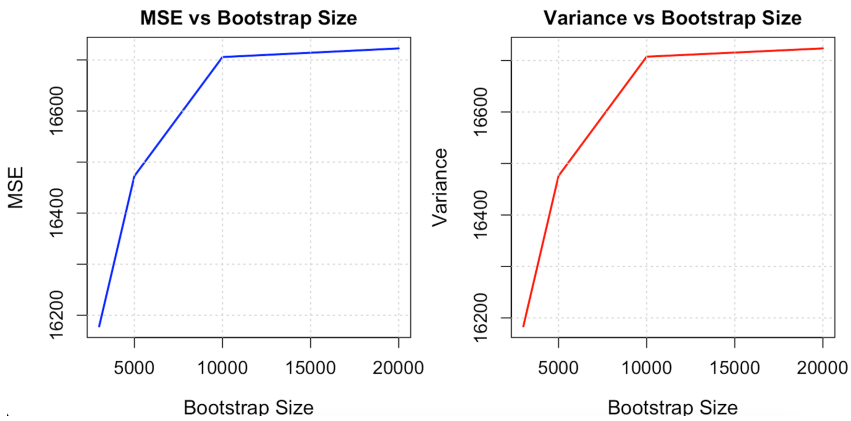

Figure 2: Mean Squared Error (MSE) and Variance (VAR) for Different Bootstrap Sample Sizes

The final bootstrap estimate of the pure premium after computation is given by:

$$\textbf{Bootstrap Estimate of Pure Premium:} \quad \textbf{843.5798} \tag{2}$$

The histogram in Figure(3) illustrates the distribution of bootstrap estimates. A red vertical line represents the mean of the bootstrap estimates.

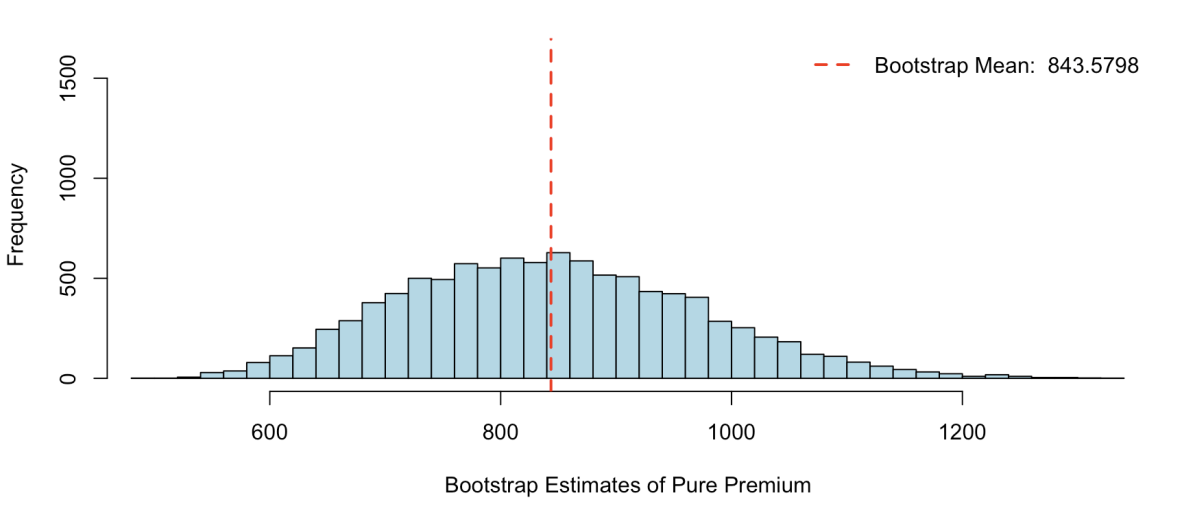

Figure 3: the distribution of bootstrap estimates

The non-parametric bootstrap method provides a more robust estimate by resampling the data multiple times. Compared to Task A1, the bootstrap estimate of pure premium **843.5798** is slightly lower than the previous estimate **856.9033**, due to resampling variability. Additionally, the bootstrap approach quantifies estimation uncertainty, offering a deeper insight into the data distribution. Increasing bootstrap repetitions reduced bias (MSE - VAR), leading to more stable estimates.

4 Task A3: Parametric Method (MLE) for Pure Premium Estimation

Maximum Likelihood Estimation (MLE) is a parametric method for estimating the parameters of a statistical model by maximizing the likelihood function. In this report, we apply MLE to fit various parametric distributions to claim size data to estimate the pure premium.

We chose five parametric distributions: **Log-Normal**, **Exponential**, **Gamma**, **Weibull**, and **Pareto**. These distributions were selected for their ability to model skewed and heavy-tailed data, which are common characteristics of claims data.

The MLE method was implemented using the `fitdist` function from the fitdistrplus R package. This function fits the selected distributions to the data by maximizing the likelihood function. We fitted five distributions and compared them using the **Akaike Information Criterion (AIC)** and **Bayesian Information Criterion (BIC)**. The LogNormal distribution had the lowest **AIC (6948)** and **BIC (6957)**, indicating it provided the best fit. The comparison of the distributions based on AIC and BIC is shown in the table (3).

No.	**Distribution**	**AIC**	**BIC**
1	LogNormal	6948	6957
5	Pareto	6959	6968
4	Weibull	7006	7014
3	Gamma	7080	7088
2	Exponential	7318	7322

Table 3: Comparison of AIC and BIC for different distributions

We also present the Q-Q plot in Figure (4), where the **Log-Normal** distribution appears to fit the data well, as most points lie close to the diagonal. However, slight deviations in the tails suggest that the Log-Normal distribution may not fully account for the extreme values. For the other distributions, we observe the following:

- **Exponential** shows significant deviations, particularly in the tails, indicating a poor fit.
- **Gamma** exhibits a moderate fit, with some deviations noticeable in the upper tail.
- **Weibull** provides a reasonable fit, although minor deviations in the tails are observed.
- **Pareto** demonstrates significant deviations, especially in the upper tail, suggesting a poor fit.

The maximum likelihood estimation (MLE) method was applied to estimate the parameters of the Log-Normal distribution. The estimated parameters are shown in table (4).

Parameter	Estimated Value
Mean of the logarithm (meanlog)	4.999351
Standard deviation of the logarithm (sdlog)	1.692362

Table 4: Estimated Log-Normal Parameters

Using these estimates, the following metrics were calculated for the Log-Normal distribution as shown in table (5):

Metric	Calculated Value
Mean	621.0405
Standard deviation	2525.207
90th percentile	1297.508

Table 5: Calculated Log-Normal Distribution Metrics

Next, the pure premium was determined using the following formula(1). Using the computed values, we have:

Non-Parametric Estimate of Pure Premium: 1103.34

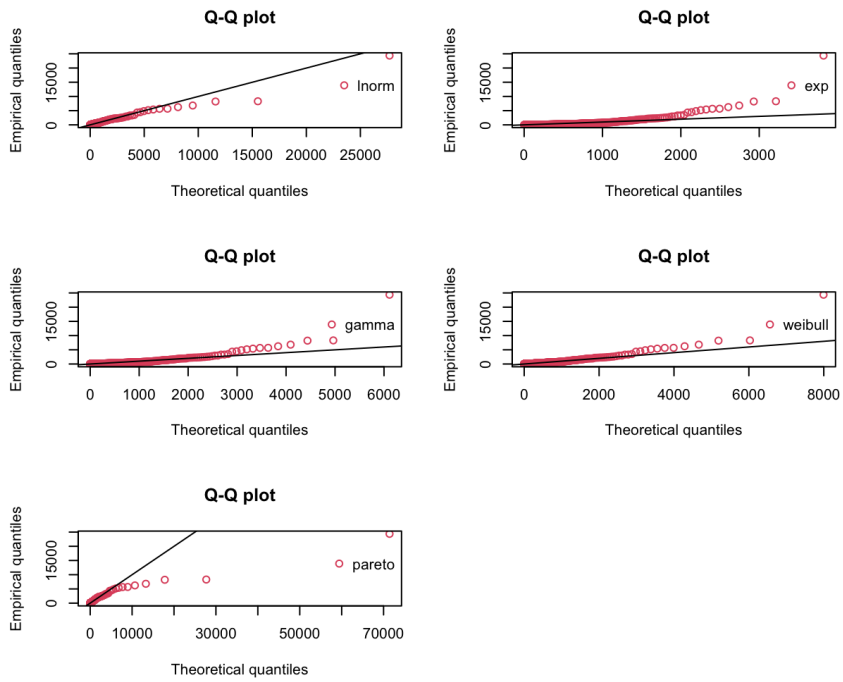

Figure 4: Q-Q plot for all five distribution

5 Task B1: Combined Data Analysis (New Dataset)

In this task, we extend our estimation of the pure premium by incorporating additional claim data `extradataset7`. The histogram below is plotted to visually analyze the data.

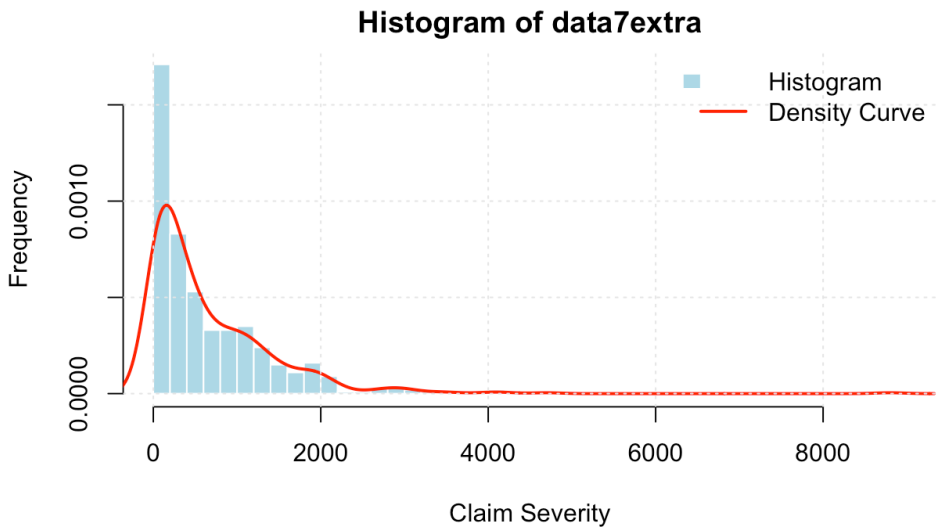

Figure 5: Histogram for extradata7

The objective was to assess whether the two datasets, `dataset7` and `extradataset7`, originate from the same distribution, and subsequently use both non-parametric and parametric estimation methods to calculate the pure premium. The **Anderson-Darling test** was employed to evaluate the similarity. As we have observed some ties in our data, to overcome these ties or repeated claims, we refer to the Anderson-Darling test to the Kolmogorov–Smirnov test. Anderson-Darling test is sensitive to deviations in the tails. The results are summarized in the table below:

Version	AD Criterion	T.AD	Asymptotic P-value
Version 1	28.768	36.540	4.7330×10^{-16}
Version 2	28.800	36.554	5.1114×10^{-16}

Table 6: Anderson-Darling Test Results

Since the p-value is below 0.05, we reject the null hypothesis, confirming that the two datasets have **different distributions**.

We redo Task A1 and A2 as follows: In Task A1, the pure premium for the extra dataset was calculated using the formula (1). The result was 842.3662. In Task A2, bootstrap resampling with 10,000 iterations was performed for bothdatasets, *data7* and *data7extra*. Each bootstrap sample was used to compute the pure premium for both datasets, and the final estimate was derived as the average of all bootstrap iterations, resulting in 841.6611. The final results are displayed as shown in the table below.

Method	Estimated Pure Premium
Using the original formula for extra data	842.3662
Using the original formula	849.6348
Using bootstrap resampling	841.6611

Table 7: Estimated Pure Premium Values

Additionally, a histogram of bootstrap estimates was generated to visualise the distribution of estimated pure premium values.

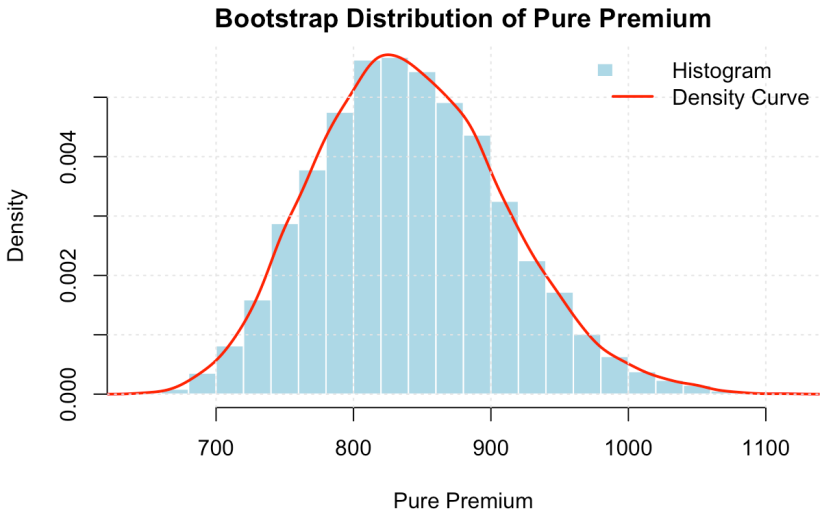

Figure 6: Bootstrap Distribution of pure premuim with combined data

We use Maximum Likelihood Estimation (MLE) to fit parametric distributions to the additional dataset (`data7extra`) and identify the best-fitting distribution. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values were compared to evaluate the fit, with lower AIC or BIC values indicating a better fit. This helps refine the pure premium estimation.

Distribution	AIC	BIC
Gamma	7470.550	7478.979
Weibull	7474.073	7482.502
Pareto	7499.485	7507.914
Exponential	7513.303	7517.518
LogNormal	7595.493	7603.922

Table 8: Comparison of AIC and BIC values for different distributions.

Based on the AIC and BIC result, we might choose the **Gamma distribution** as the best-fitted model for this extra dataset.

To visually assess the goodness of fit for each parametric distribution, Quantile-Quantile (Q-Q) plots were generated. These plots compare the quantiles of the observed claims data against the quantiles of the fitted theoretical distributions.

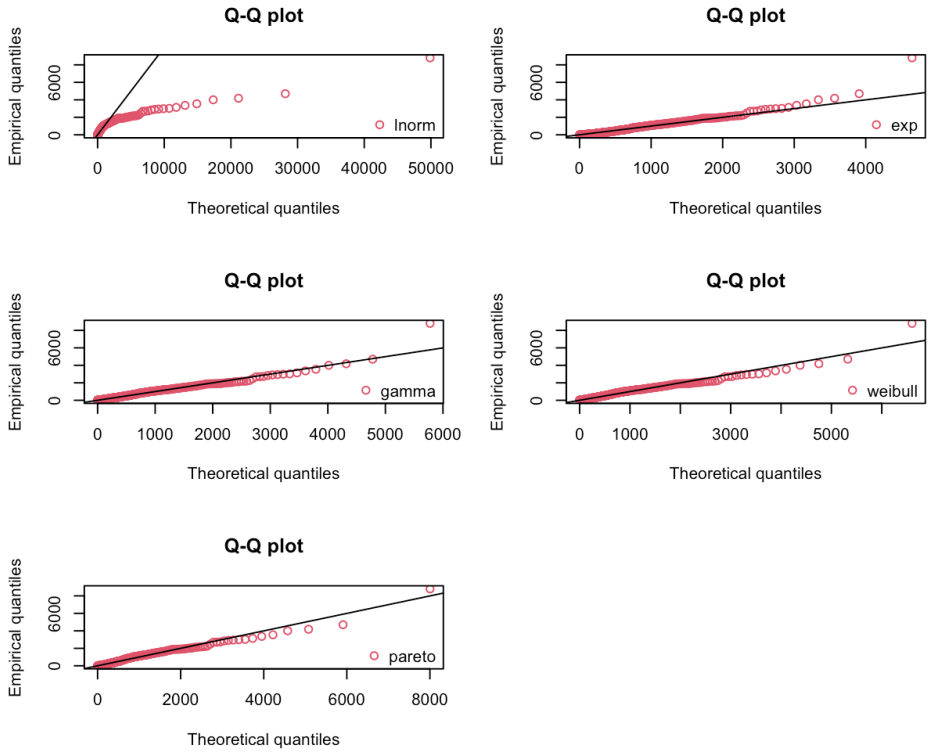

Figure 7: Q-Q plot for best fit distribution for extradata7

The parameters of the Gamma distribution were estimated using Maximum Likelihood Estimation (MLE). The shape parameter was found to be **0.4849**, and the rate parameter was **0.000877**. With these parameters, the mean, standard deviation, and 90th percentile of the Gamma distribution were calculated. Using these values, the pure premium for the additional dataset (`data7extra`) was estimated to be **744.0482**.

Finally, the pure premium estimate obtained via a parametric approach was derived by averaging the pure premiums from both the Gamma and Log-Normal distributions, resulting in a final estimate of **923.6961**.

Method	**Estimated Pure Premium**
Gamma Distribution (extra data)	744.0482
Final Parametric Estimate	923.6961

Table 9: Estimated Pure Premium Values Using the Gamma Distribution

6 Task B2: Non-Parametric Method for Aggregate Loss Distribution (2025 Portfolio)

The insurer has decided to underwrite 5,000 policies in 2025. Each policy has a probability of 0.6 of generating exactly one claim and a probability of 0.4 of generating no claims. A non-parametric method is used to estimate the aggregate loss distribution of the insurer's portfolio by simulating losses over 100,000 iterations. The resulting distribution provides insights into the overall risk exposure of the insurer.

Each policy follows a Bernoulli distribution with a probability of 0.6 for generating a claim and 0.4 for no claim:

$$X_i \sim \text{Bernoulli}(0.6) \quad \text{for} \quad i = 1, 2, \ldots, n$$

where X_i represents the number of claims for policy i.

For each policy that results in a claim, the severity amount is sampled from either `data7` or `data7extra` with equal probability:

$$S_i = \begin{cases} \text{data7} & \text{with probability } 0.5, \\ \text{data7extra} & \text{with probability } 0.5. \end{cases}$$

where S_i is the claim amount.

The total aggregate loss for the insurer's portfolio is computed as:

$$L_{\text{total}} = \sum_{i=1}^{n} X_i \cdot S_i$$

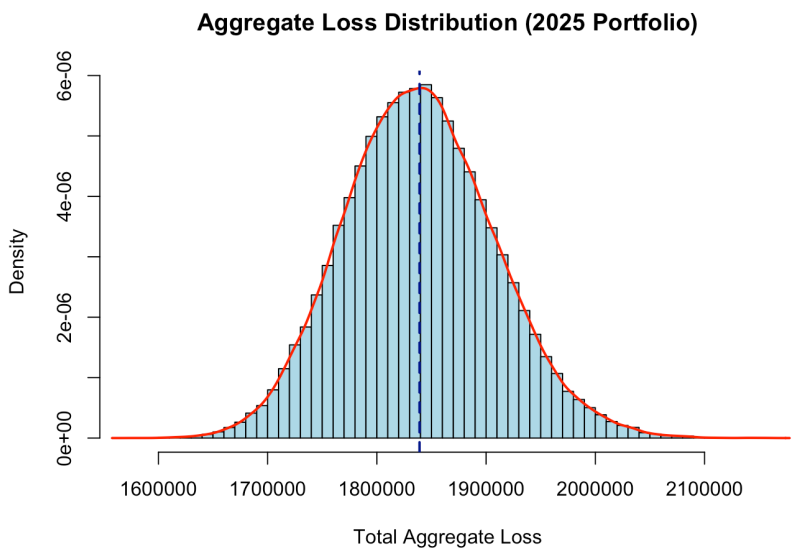

Figure 8: Histogram for agreggate losses

This process is repeated 100,000 times using Monte Carlo simulation to generate the aggregate loss distribution. After simulating, we have found that the mean aggregate loss across 100,000 iterations is **1,839,061**, indicating the expected total loss for the insurer's portfolio.

To visualise we plot histogram in figure (8) shown above. The overlaid red density curve provides a smooth approximation of the aggregate loss distribution. A dashed vertical line marks the mean value of aggregate losses, ensuring clear visual interpretation.

7 Task B3: Parametric Method for Aggregate Loss Distribution (2025 Portfolio)

In Task B3, we applied a parametric method to estimate the aggregate loss distribution for the insurer's portfolio in 2025. This method is based on the assumption that the claim severity follows specific distributions that have been fitted to the data using Maximum Likelihood Estimation (MLE) in Task B1.

In Task B1, we estimated the parameters for the **lognormal** and **gamma** distributions to model the severity of claims from two different datasets: `data7` (for regular claims) and `data7extra` (for extra claims). The estimated parameters obtained from MLE for each distribution are as follows in the table (10).

Distribution	Parameter	Estimated Value
Lognormal	meanlog	4.999351
	sdlog	1.692362
Gamma	shape	0.7094708
	rate	0.001054797

Table 10: Estimated Parameters for the Fitted Distributions

For the parametric method, we assumed that each policy has a 60% chance of generating exactly one claim (following a Bernoulli distribution with a probability of 0.6), and a 40% chance of generating no claim. Once a claim was generated, the severity amount was sampled from either the **lognormal distribution** (for regular claims) or the **gamma distribution** (for extra claims), with equal probability (50% chance each).

The total aggregate loss for the insurer's portfolio was then computed as the sum of the severities for all policies that generated claims. This process was repeated 100,000 times using Monte Carlo simulation to generate a comprehensive estimate of the aggregate loss distribution.

The simulation yielded an **average (mean)** aggregate loss of **1,940,248** for the insurer's portfolio in 2025. This value represents the expected total loss for the portfolio, based on the simulated claims and severity distributions.

Further, the aggregate losses were visualized in a histogram, which illustrated the distribution of potential outcomes, and the mean value was clearly marked on the plot. The histogram visually showed the spread of possible total losses, and the vertical line corresponding to the mean aggregate loss served to highlight the expected loss.

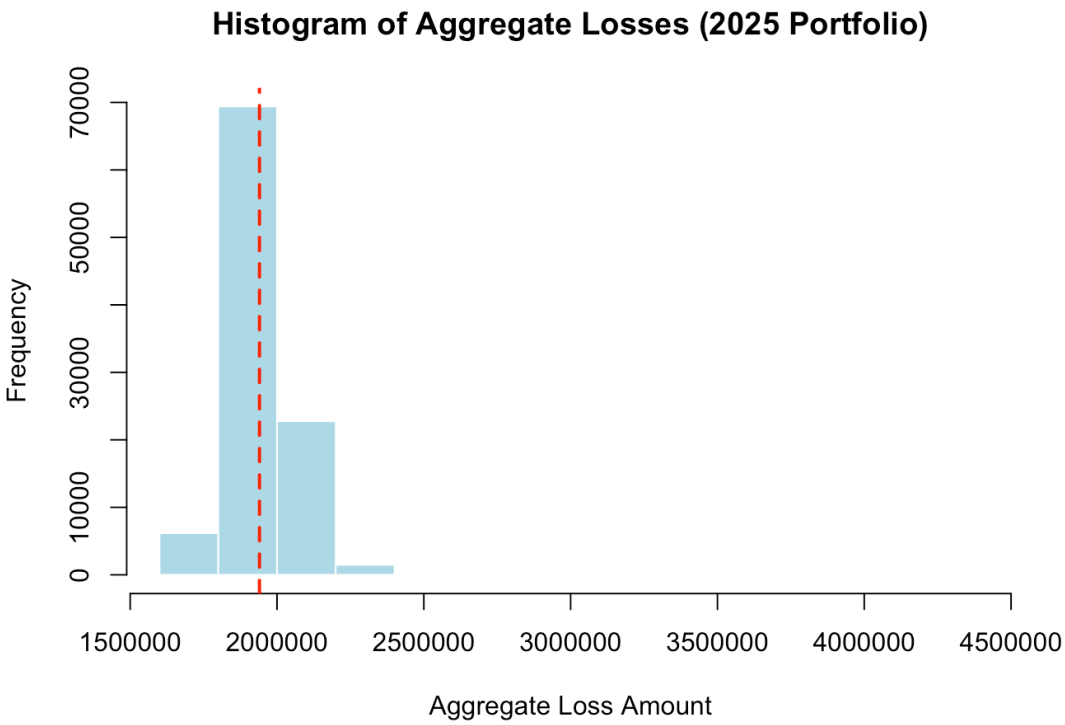
Histogram of Aggregate Losses (2025 Portfolio)
Frequency
0
10000
30000
50000
70000
1500000
2000000
2500000
3000000
3500000
4000000
4500000
Aggregate Loss Amount

Figure 9

8 Task C1: Non-Parametric Estimation with Outliers Removed (2025 Portfolio)

In this task, we reassess our aggregate loss estimation by identifying and removing outliers from the dataset. While these extreme values may be legitimate, their inclusion in aggregate loss modeling can skew estimates and exaggerate risk. Due to their high severity and low frequency, they should be analyzed separately and modeled using specialized methods to prevent excessive volatility.

We used the Interquartile Range (IQR) method to identify and remove outliers in datasets data7 and data7extra, which eleminates values exceeding three times the IQR above the third quartile or below the first. This conservative approach ensures that frequently occurring high-severity claims, indicative of a heavy-tailed distribution, are preserved.

Applying this criterion, **42 total outliers** were detected (38 outliers in data7 and 4 outliers in data7extra), representing approximately **4.2%** of total claims. Their removal reduced data dispersion, as reflected in a lower standard deviation.

After 100,000 simulations, a significant decrease in the mean aggregate loss, from **1,839,061** to **1,357,456**, while the standard deviation decreased by **34,395.64**. This suggests that outliers significantly influenced the variability of expected losses.

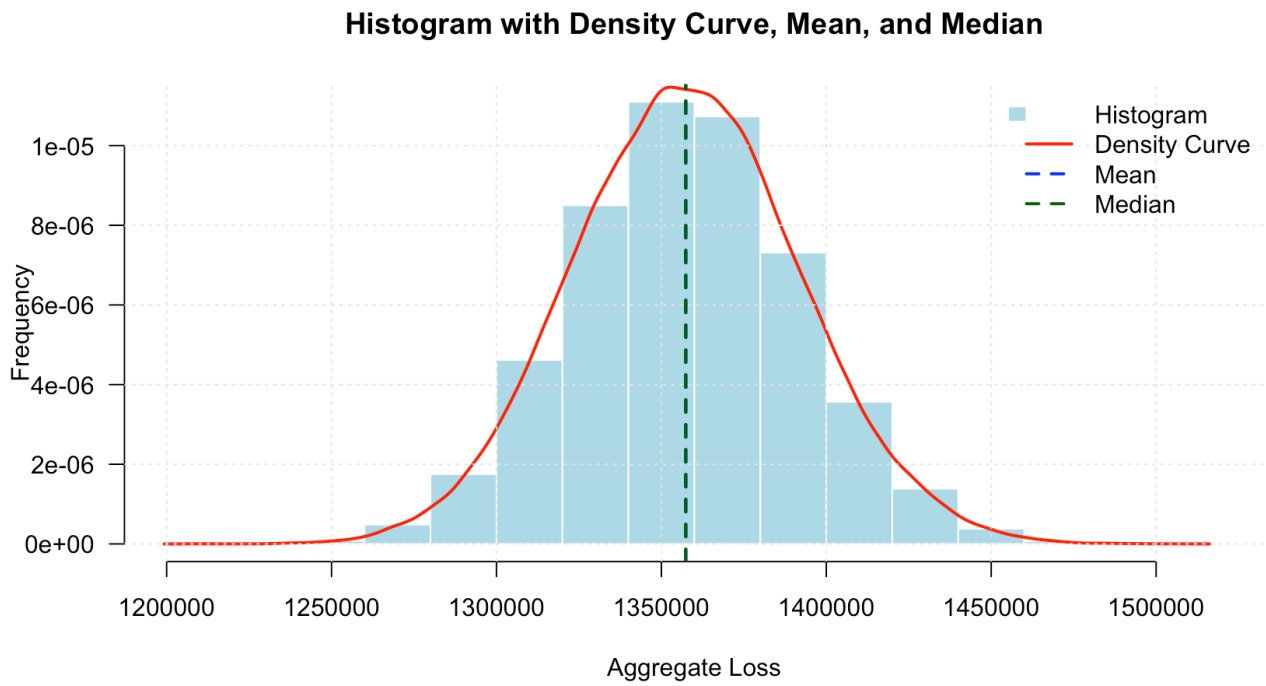

Figure 10

Table 11: Comparison of Aggregate Loss Estimates Before and After Outlier Removal

Method	**Mean Aggregate Loss**
With Outliers	1,839,061
Without Outliers	1,357,456

The histogram (figure 10) indicates that the adjusted distribution is more symmetric, with a reduced likelihood of extreme values. By eliminating outliers, the estimate becomes more stable, which is beneficial for setting reserves or crafting reinsurance strategies. In summary, this modification helps minimize volatility in loss estimation and offers a clearer understanding of the risk portfolio.

9 Task C2: Parametric Estimation with Outliers Removed (2025 Portfolio)

To assess the impact of outliers on the parametric estimation of aggregate loss, we applied the same removal criteria from C1 before fitting a parametric distribution. Extreme values can distort maximum likelihood estimators (MLE), affecting model selection and leading to biased risk estimates.

A total of 45 outliers were identified and removed, representing approximately 4.2% of claims. We refitted distributions to the filtered data and compared results with those from the original dataset. Before removing outliers, Gamma was the best fit for Data7Extra, while Log-Normal was preferred for Data7. After removing outliers, the best fit changed to Weibull (Data7) and Gamma (Data7Extra),as table 12 and 13 suggests. This confirms that extreme values significantly influenced model selection, reinforcing the need to handle outliers carefully.

The change [1]in distribution does not lead to a significant difference in aggregate loss for outliers in comparison to maintaining the same distribution as in Task B3, since outliers have a minimal impact on the aggregate loss because the weighting scheme in the pure premium formula primarily depends on the mean, standard deviation, and 90th percentile. While outliers can slightly influence the standard deviation and the 90th percentile, their impact is diluted by the dominant weight on the mean (65%). As a result, even with a change in distribution, the overall aggregate loss remains relatively stable.

Distribution	AIC	BIC
Weibull	6053.101	6061.372
Gamma	6060.624	6068.895
Pareto	6062.691	6070.962
LogNormal	6070.302	6078.573
Exponential	6101.104	6105.240

Table 12: Comparison of AIC and BIC for Data7

Distribution	AIC	BIC
Gamma	7362.408	7370.822
Weibull	7369.526	7377.939
Pareto	7394.840	7403.253
Exponential	7395.061	7399.267
LogNormal	7503.458	7511.871

Table 13: Comparison of AIC and BIC for Data7extra

[1]We could also consider the LogNormal and Gamma distributions as potential fits, as outlined in the R code documentation. Adjusting the distribution may lead to more precise predictions and better model performance, especially when outliers or other factors influence the results.

Parameter	Value
Weibull	
Meanlog (data7)	4.745798
SDlog (data7)	1.492609
Gamma	
Shape (data7extra)	0.8333686
Scale (data7extra)	580.3409856

Table 14: Summary of Parameter Estimates for Weibull and Gamma Distributions

From table (14), we observe that the mean and standard deviation decreased for Log-Normal, suggesting reduced data dispersion. The Weibull shape and scale parameters indicate a slightly flatter distribution compared to the original dataset.

After 100,000 simulations using the adjusted distribution, the mean aggregate loss decreased from **1,940,248** to **1,477,197**, a **24%** reduction, reflecting a more stable estimate. The standard deviation also decreased from **104,722.18** to **51,812.92**, a **50%** reduction, indicating that outliers were significantly driving the variability. Removing these extreme values resulted in a less skewed and more reliable loss distribution.

The histogram of the new parametric estimate shows a less skewed distribution with reduced impact from extreme values. Compared to the original estimate, this refined version may provide a more stable risk assessment, improving decision-making for reserve setting and reinsurance strategies.

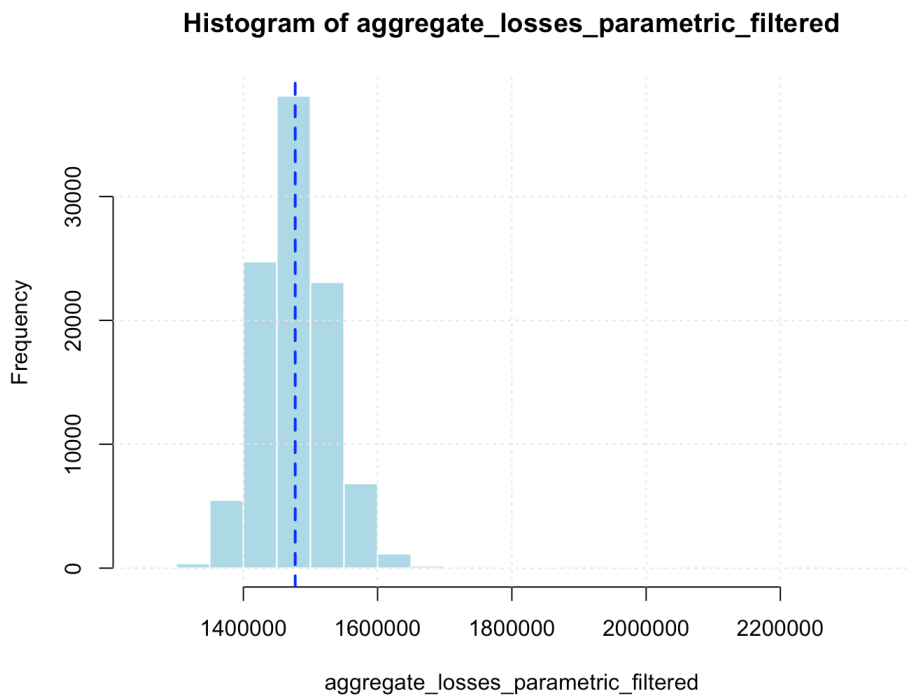

Figure 11